

EASTERN WEST VIRGINIA, WV, USA

WMO: 724177

Lat: 39.402N

Lon: 77.984W

Elev: 534

StdP: 14.41

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13734

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	15.4	-4.9	4.3	15.4	-0.2	5.5	18.6	27.6	35.0	25.2	33.1	7.0	310	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	92.4	73.8	89.9	73.0	87.4	72.0	76.8	87.5	75.4	85.3	74.2	83.5	8.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	126.8	81.0	72.4	122.4	79.9	71.4	118.1	78.8	40.4	87.4	39.1	85.4	37.9	83.5	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
22.9	19.1	16.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			2.7	97.1	5.9	3.5	-1.5	99.7	-4.9	101.7	-8.2	103.7	-12.5	106.3
			1.6	79.4	5.5	2.0	-2.3	80.9	-5.6	82.0	-8.7	83.1	-12.7	84.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	32.5	35.3	43.2	54.3	63.0	71.6	75.7	74.0	67.2	55.8	45.2	36.6	(6)
(7)		DBStd	17.09	10.60	9.58	9.50	8.50	7.73	5.81	4.93	4.73	6.64	7.99	8.41	8.70	(7)
(8)		HDD50	1929	553	421	254	48	3	0	0	0	31	192	427	(8)	
(9)		HDD65	4830	1008	832	676	335	134	13	0	1	52	303	595	881	(9)
(10)		CDD50	3622	10	9	45	179	405	648	796	744	517	211	47	11	(10)
(11)		CDD65	1050	0	0	2	15	71	211	332	281	119	19	0	0	(11)
(12)		CDH74	9787	0	5	27	228	731	1936	3184	2468	1022	178	7	1	(12)
(13)		CDH80	3536	0	0	4	57	216	675	1318	894	335	37	0	0	(13)
(14)	Wind	WSAvg	6.0	7.4	7.6	7.9	7.4	5.6	5.1	4.6	4.1	4.5	5.3	6.2	6.4	(14)
(15)	Precipitation	PrecAvg	38.1	2.5	2.4	3.4	3.1	4.0	3.6	3.6	3.2	3.3	3.1	3.0	2.6	(15)
(16)		PrecMax	60.2	6.3	5.8	7.8	7.8	9.7	9.2	8.7	7.2	9.4	9.5	7.8	6.4	(16)
(17)		PrecMin	26.3	0.5	0.2	0.3	0.2	0.4	0.7	0.4	0.6	0.2	0.1	0.3	0.5	(17)
(18)		PrecStd	7.3	1.4	1.5	1.6	1.7	2.3	2.0	1.9	1.7	2.5	2.1	1.7	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.2	70.1	78.7	86.2	90.0	94.0	97.5	93.8	92.3	84.4	74.0	68.4	(19)
(20)		MCWB	57.7	56.9	60.3	64.1	70.9	74.7	74.8	73.7	72.0	70.0	59.6	58.3	(20)	
(21)		2%	DB	60.5	61.8	70.9	80.6	85.7	90.0	93.2	90.5	87.9	79.1	68.4	60.8	(21)
(22)		MCWB	53.5	50.5	56.1	62.2	69.6	73.4	74.4	73.8	70.3	66.0	57.3	53.1	(22)	
(23)		5%	DB	53.9	55.2	65.1	75.3	81.9	87.0	90.2	87.9	83.8	74.3	64.0	54.6	(23)
(24)		MCWB	47.5	45.8	53.7	60.0	67.1	72.2	73.5	72.7	69.3	63.1	55.2	48.2	(24)	
(25)		10%	DB	47.4	49.9	60.2	70.5	77.8	84.0	87.3	85.2	80.1	70.3	59.6	49.8	(25)
(26)		MCWB	41.6	42.3	50.1	57.0	65.3	70.4	72.5	71.4	67.5	61.2	51.2	43.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	59.2	62.7	67.8	74.1	77.7	78.5	78.3	76.0	72.3	63.6	60.4	(27)
(28)		MDCB	64.5	67.5	75.8	79.8	85.5	88.3	91.1	88.7	84.7	79.8	69.3	66.2	(28)	
(29)		2%	WB	55.0	51.9	58.9	65.0	71.5	75.4	77.0	76.4	73.7	69.2	60.2	54.7	(29)
(30)		MDCB	59.4	59.2	67.4	76.3	81.5	86.0	88.6	86.0	81.5	76.0	66.0	59.1	(30)	
(31)		5%	WB	47.7	47.0	55.1	62.0	69.4	73.9	75.5	74.9	72.1	65.9	56.6	49.2	(31)
(32)		MDCB	53.7	53.5	63.8	72.1	77.9	84.0	86.1	83.7	79.4	72.0	62.2	53.9	(32)	
(33)		10%	WB	42.0	42.9	50.8	59.2	67.3	72.4	74.2	73.4	70.2	62.5	52.6	44.3	(33)
(34)		MDCB	46.8	49.4	59.0	68.5	75.4	81.3	83.9	81.6	76.7	68.7	58.2	49.0	(34)	
(35)	Mean Daily Temperature Range		MDBR	15.9	17.5	19.6	22.1	21.3	20.6	20.6	20.4	21.0	21.0	19.2	16.0	(35)
(36)		5% DB	MCDBR	23.9	27.0	28.3	29.8	27.0	24.3	24.3	23.5	26.2	27.2	26.6	23.3	(36)
(37)		MCWBR	17.8	18.5	17.5	15.5	13.1	10.8	8.8	9.5	11.2	15.0	17.6	17.0	(37)	
(38)		5% WB	MCDBR	22.8	24.0	25.7	25.1	23.1	21.4	20.6	19.9	20.0	22.1	21.7	20.6	(38)
(39)		MCWBR	18.1	17.5	17.5	14.7	12.1	10.2	8.7	9.1	10.0	13.4	16.6	16.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.324	0.357	0.390	0.428	0.495	0.514	0.511	0.428	0.368	0.323	0.309	(40)	
(41)		taud	2.450	2.379	2.364	2.301	2.219	2.065	2.031	2.022	2.233	2.401	2.490	2.477	(41)	
(42)		Ebn at Noon	274	284	285	281	271	252	246	243	258	265	266	265	(42)	
(43)		Edh at Noon	25	31	36	40	45	52	54	53	40	30	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	610	884	1181	1512	1685	1869	1836	1642	1344	990	706	531	(44)	
(45)		RadStd	54	108	68	125	171	143	106	87	173	114	63	53	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.72	N/A	N/A	N/A	N/A	(46)
(47)	Regional (61 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+113	(47)

Nomenclature: See separate page